Anna Bellach, Ph.D.

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abellach.github.io

Biostatistician | Expertise in Survival Analysis • Competing Risks • Semiparametric Modeling • Transfer Learning • Mathematical Statistics • Empirical Processes • Clinical Trials

Professional Experience

2025 – present	Visiting Research Scholar at the University of North Carolina, Chapel Hill
2025	Senior Researcher at the German Cancer Research Center, Heidelberg
2021 – 2024	Mathematical Statistician at the National Heart, Lung, and Blood Institute of the US National Institutes of Health (contractor)
2017 – 2021	Postdoctoral Researcher at Fred Hutch Cancer Center, University of Washington, and Yale School of Public Health
2012 – 2017	Predoctoral Researcher at the University of Copenhagen,
2014, 2015	Visiting Research Scholar at the University of North Carolina, Chapel Hill
2012	Research Assistant at University of Copenhagen
2004 – 2009	Instructor Teaching Assistant at Albert Ludwig University of Freiburg

Education

2017	PhD, Biostatistics and Bioinformatics, University of Copenhagen, Denmark
2012	MSc, Mathematics (Diplom), Albert Ludwig University of Freiburg, Germany
2004	BSc, Mathematics (Vordiplom), Albert Ludwig University of Freiburg, Germany

Awards

2018	Summer Institutes Scholarship, University of Washington
2018	SPAC Course Scholarship, Fred Hutch Cancer Center
2012 – 2017	Marie-Curie PhD fellow, initial training network MEDIASRES funded by the European Commission under grant agreement no. 290025

Peer Reviewed Publications:

A. Methodological Research:

1. **Bellach A**, Kosorok MR, Gilbert PB & Fine JP (2020). General regression model for the subdistribution of a competing risk under left-truncation and right-censoring. *Biometrika* 107(4): 949-964, DOI:10.1093/biomet/asaa034.
2. **Bellach A**, Kosorok MR, Rüschendorf L & Fine JP (2019).Weighted NPMLE for the subdistribution of a competing risk. *Journal of the American Statistical Association* 114:259-270, DOI:10.1080/01621459.2017.1401540.

B. Collaborative Research

3. Sorror ML, Saber W, Brent RL, Geller N, **Bellach A**, et al. (2026). CHARM is Prognostic of Geriatric Morbidity and Toxicity after Allogeneic Transplant for Older Adults: BMT CTN 1704 Study. *Blood Advances*, 10 (5): 1657-1669, DOI:10.1182/bloodadvances.2025018340.
4. Sorror ML, Logan B, Saber W, Geller N, **Bellach A**, et al. (2025). Novel Composite Health Assessment Risk Model for Older Allogeneic Transplant Recipients: BMT-CTN 1704. *Blood Advances*, 9 (13): 3268-3280, DOI:10.1182/bloodadvances.2025015793.
5. Cathey B, **Bellach A**, Troendle J, Smith K, Raja N, Osgood S, Kozel B & Levin M (2024). Increased heart rate fragmentation in those with Williams-Beuren syndrome suggests non-autonomic mechanistic contributors to sudden death risk. *AJP-Heart and Circulatory Physiology*, 327(2): H521–H532, DOI:10.1152/ajpheart.00601.2023.

Submitted Manuscripts

6. **Bellach A** & Kosorok MR. General regression model for recurrent event data with competing terminal events.

Manuscripts in Preparation

7. Sharma A, Mangaonkar AA, Kassim AA, Fritz AR, **Bellach A**, et al. (2024). Trial in Progress: Cureaa-a Clinical Trial of Upfront Haploidentical or Unrelated Donor BMT to Restore Normal Hematopoiesis in Aplastic Anemia, Blood and Marrow Transplant Clinical Trials Network BMT CTN 2207. *Blood*, DOI:10.1182/blood-2024-207952.

Teaching Experience - Teaching Assistant at University of Freiburg, Germany

- Probability Theory I–II / Mathematical Statistics
- Stochastic Processes
- Financial Mathematics
- Higher Mathematics for Engineers
- Independently organized and taught summer review courses to prepare students for Bachelor's examinations in Probability and Statistics

Selected Projects at the Office of Biostatistics Research/NHLBI

1. **BMT CTN 1704:** *Novel composite health assessment risk model for older allogeneic transplant recipients.* Blinded Statistician.
2. **BMT CTN 2207:** *Clinical Trial of Upfront Haploidentical or Unrelated Donor BMT to Restore Normal Hematopoiesis in Aplastic Anemia.* Blinded Statistician.
3. **BMT CTN 2303:** *A Randomized Phase 3 Trial of Remestemcel-L and Ruxolitinib for Steroid-Refractory Acute Graft-versus-Host Disease.* Blinded Statistician.
4. **WARRIORS Trial:** *Women's Aneurysm Research: Repair Immediately Or Routine Surveillance.* Blinded Statistician (Extramural).
5. **TransIT:** *Phase III Randomized Trial: URD Bone Marrow Transplantation vs. IST for Pediatric Severe Aplastic Anemia.* Unblinded Statistician.
6. **Heart Rate Fragmentation** *in Children and Adolescent with Williams Syndrome.* Project Statistician.
7. **Wearable Device Study** *on Sleeping Patterns in Williams Syndrome.* Project Statistician.
8. **Protocol Reviews:** Statistical review of various protocols for DIR clinical trials.

Invited Seminar Talks

1. *Semiparametric regression models for recurrent and terminal events,* St.Jude Children's Research Hospital, Memphis, August 2025.
2. *Semiparametric regression models for recurrent and terminal events,* UCLA (online), July 2025.
3. *Semiparametric regression models for recurrent and terminal events,* US National Cancer Institute, Bethesda, November 2023.
4. *Competing risks regression models based on pseudo risk sets*, Karolinska Institute, Stockholm (online), October 2021.
5. *Competing risks regression models based on pseudo risk sets*, University of Florida, Gainesville (online), September 2021.
6. *Competing risks regression models based on pseudo risk sets*, US National Heart, Lung and Blood Institute, Bethesda (online), July 2021.
7. *Competing risks regression models based on pseudo risk sets*, CBAR at Harvard T.H. Chan School of Public Health, Boston (online), August 2020.
8. *Competing risks regression models based on pseudo risk sets*, Fred Hutchinson Cancer Research Center, Seattle, October 2018.

Contributed Talks

9. *General regression model for competing risks and recurrent event cancer data subject to competing terminal events*, US National Cancer Institute, Bethesda, May 2016.
10. *General regression model for the marginal mean intensity of a recurrent event with competing terminal events*, JSM in Washingon DC, August 2022.
11. *General regression model for the subdistribution of a competing risk under left-truncation and right-censoring*, ENAR spring meeting in Philadelphia, June 2019.
12. *General regression model for the subdistribution of a competing risk under left-truncation and right-censoring*, JSM in Vancouver, July 2018.
13. *General regression model for the subdistribution of a competing risk under left-truncation and right-censoring*, ISNPS in Salerno, June 2018.
14. *Competing risks regression models based on pseudo risk sets*, ENAR spring meeting in Washington, D.C., March 2017.
15. *Competing risks regression models based on pseudo risk sets*, MEDIASRES final meeting, Albert-Ludwigs University, Freiburg im Breisgau, Germany, October 2015.
16. *Extending Fine and Gray's model: A New Approach to Competing Risks Analysis*, ENAR spring meeting in Miami, March 2015.
17. *Extending Fine and Gray's model: A New Approach to Competing Risks Analysis*, JSM in Boston, August 2014.

Full list of talks: abellach.github.io/talks

Referee Service

1. Journal of the American Statistical Association
2. Journal of the Royal Statistical Society: Series B
3. Nature
4. Biometrics
5. Annals of Applied Statistics
6. Lifetime Data Analysis
7. International Journal of Biostatistics
8. Statistics in Medicine
9. Statistical Methods in Medical Research
10. Statistics and Probability Letters
11. Statistics in Biosciences
12. Communications for Statistical Applications and Methods
13. Cell Reports Medicine

Programming Skills

R (advanced), LATEX (advanced), SAS (intermediate)

Professional Development

- Bridge to Leadership Program, National Institutes of Health (NHLBI)
- Deep Learning Specialization, Coursera / DeepLearning.AI
- Key Topics in Causal Inference, Harvard T.H. Chan School of Public Health
- Good Clinical Practice (GCP) Certification, CITI Program
- Human Subjects Research (HSP) Certification, CITI Program

Languages

English (fluent), German (native speaker)